Phys 224 Problem Set 5

Due Monday Nov. 16 at 11:59 PM. You may talk/consult with whom or whatever, but you must write your own solutions for submission.

1) Modes the easy way. We'll estimate the number of modes in a room up to some frequency ν_{cut}, but this time we'll do it with dimensional analysis. If we double the size of the room in one direction, we reduce the fundamental frequency by a factor of 2, so for a fixed ν_{cut}, the number of modes in that direction must go up by a factor of 2. The same must be true of the other directions, so the number of modes must be proportional to $xyz = V$ where x, y, z are the dimensions of the room and V is the volume. The number of modes is dimensionless, and besides the volume it can depend on the frequency ν_{cut} and the speed of sound. Use dimensional analysis to estimate the number of modes below ν_{cut} in a room of volume V. The true answer is a factor of $\pi/6$ smaller than this, which is very close to a half.

2) Transition frequency in OSM hall. Problem 1 tells us the number of modes depends on the volume, as long as ν_{cut} is well above the resonant frequencies. In particular, we don't have to worry about the details of the shape of a room, just its overall volume.
A) Estimate the volume of the OSM hall (A range of answers is acceptable).
B) The reverb time τ of the hall is about 2 seconds. Resonances will overlap if their frequencies are within $\frac{1}{\tau}$ of each other, and stop acting like distinct resonances. The frequencies have to start overlapping if the number of modes up to ν_{cut} is larger than the number of non-overlapping possible frequencies. I.e. if we have a τ of 2 and a max frequency of 200 Hz, then if we have more than 400 modes below 200 Hz, they have to overlap. By setting the number of modes up to ν_{cut} equal to the number of distinct slots up to ν_{cut}, which is $\tau\nu_{cut}$, estimate the frequency at which the OSM hall transitions from a set of resonant frequencies to a flat frequency response.

3) Q of the hall. Estimate the quality factor Q of the OSM at the transition frequency. As a reminder, we're defining Q to be the number of cycles during which the amplitude decays by a factor of $1/e$, or $\tau\nu$.

4) Q again. Estimate the quality factor of the spring we saw in class. If you weren't there, the spring was still moving after a few minutes, but the amplitude had noticeably decreased. When I counted, the spring went up and down about 80 times per minute.

5) Resonant driving. Assume the spring in the previous problem had a resonant frequency of exactly 80 cycles per minute. If we attach a motor that moves up and down by 0.1mm to the end of the spring, estimate the largest amplitude of the motion the mass at the end of the spring could have. To get this large response, what frequency *in Hz* should we set the motor to run, and what is the allowed error on that frequency?